



Having explored ns-3 for the past five months in *OSFY*, we now have come to the concluding article in the series. Through the series, awareness of ns-3, installation, its various tools and, finally, the traps and pitfalls associated with it have been covered. No study can ever encompass the full range of ns-3's potential. We hope some of our readers will now be motivated enough to seek more knowledge about ns-3 and apply it.

This is the last article in the series on ns-3 and I would like to use Winston Churchill's quote from an entirely different context: "Now this is not the end. It is not even the beginning of the end. But it is, perhaps, the end of the beginning." He was referring to the second battle of El Alamein, Egypt, when the Allied army defeated the Axis powers. Yes, the series is ending but our endeavour to master ns-3 continues. Over the past issues, I tried to give you an overview of ns-3. This tutorial is not complete, but the material I have covered is sufficient to initiate you into the intricacies of ns-3. Before we wind up, let's try to tie up a few more loose ends.

Python in ns-3

I have mentioned that the use of Python is not mandatory in ns-3. But I need to explain the significance of Python in ns-3, because those who are familiar with Tcl in ns-2 expect a similar role for Python in ns-3. This is a misconception. The ns-3 and ns-2 simulators are independent of each other and

the way Python interacts with ns-3 is entirely different from the way Tcl interacts with ns-2.

So what is the relationship between Python and ns-3? Are they mere acquaintances, friends or brothers? A technique called Python bindings is used in ns-3 to write ns-3 simulation scripts in Python. Python binding is a general technique used to access libraries in other programming languages from Python. For example, it is possible to create a binding to access C++ libraries from Python. But we already know how to simulate ns-3 scripts directly using C++. Then why is a second language like Python used to replicate the same results? Well, right now, that's all Python is offering. The interaction between Python and ns-3 is still a work in progress. But when Python bindings in ns-3 are complete, as predicted, we will have the ability to create new prototype models in ns-3, using Python alone. For example, we might have the ability to simulate a new routing protocol using Python without modifying the C++ source files of ns-3. But until then, the use of Python in ns-3 is not highly recommended for beginners.